Lecture 5 - Properties of Fields

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From our prior definition of a field, we can construct a number of useful theorems.

Theorem 1. *The addition-cancellation property:* $x + y = x + z \implies y = z$.

Proof.

$$\begin{aligned} y &= 0 + y \\ &= (-x + x) + y \\ &= -x + (x + y) \\ &= -x + (x + z) \\ &= (-x + x) + z \\ &= 0 + z \\ &= z \end{aligned}$$

□

Corollary 1.1. *The additive inverse of any element in a field is unique*

Proof. For the purposes of contradiction, assume that for $x \in \mathbb{F}$, there exists two distinct additive inverses $-x_1, -x_2 \in \mathbb{F}$, $-x_1 - x_2 = 0$ of x. Then, we have:

$$\begin{aligned} -x_1 + x &= 0 \\ &= -x_2 + x \end{aligned}$$

Which, by the addition-cancellation property, implies that $-x_1 = -x_2$, and thus, by contradiction, the additive inverse of an element in $\mathbb{F}$ is always unique □

Proof. We can prove this by considering the product of any number $x \in \mathbb{F}$ with 0.

$$\begin{aligned} 0 \cdot x + 0 \cdot y &= (0 + 0) \cdot x \\ &= 0 + \cdot x \end{aligned}$$

Therefore, by the addition-cancellation property, $0 \cdot x = 0 \; \forall x \in \mathbb{F}$. □

1 Example: Prove that the product of anything with zero is always zero